First Order System

Pendulum

$\frac{dx_1}{dt} = x_2$

$\frac{dx_2}{dt} = -\frac{g}{L} sin(x_1)$

What it Models?

Pendulum

The pendulum equation is a way to model a real world pendulum consisting of a weight and massless cord to create a pivot, such as in a simple gravity pendulum. These equations can be used to find the length of cord, gravitational field, and Θ or the angle the pendulum lies at at a given time. x_1 and x_2 each respectively represent Angular Position(1) and Angular Velocity(2).

Parameters

Pendulum

Our parameters for the Pendulum equation are our gravitational field (g) and the length of our chord (L).

Sources

Wikipedia. Pendulum (mechanics). https://en.wikipedia.org/wiki/Pendulum_(mechanics)